

Seyed Alireza Mousavian Anaraki

PhD in Data Science, University of Rome Tor Vergata | Visiting Researcher, University of Sheffield

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EDUCATION

PhD Student in Data Science

Nov. 2023 – Oct. 2026 (Expected*)

University of Rome Tor Vergata, Department of Enterprise Engineering

Rome, Italy

- **Research Focus: NLP for sustainability (ESG) analysis**, large-scale processing of sustainability reports, semantic alignment and weak-supervision techniques, and LLMs for structure, consistency, and credibility assessment
- *Thesis defense scheduled for April–May 2027
- **Supervisors:** Profs. Danilo Croce (croce@dii.uniroma2.it) and Roberto Basili (basili@dii.uniroma2.it)

Master's Degree in Engineering Management

Sept. 2019 – Nov. 2021

Iran University of Science and Technology, Department of Industrial Engineering

Tehran, Iran

- **GPA: 19.33/20 — Ranked 1st** student in Master's program
- **Thesis:** Development and implementation of data mining methods by creating strong, balanced, and interpretable clusters to adopt appropriate policies
- **Supervisor:** Dr. Abdorrahman Haeri (ahaeri@iust.ac.ir)

Bachelor's Degree in Industrial Engineering

Sept. 2015 – Aug. 2019

Kashan University, Department of Industrial Engineering

Kashan, Iran

- **GPA: 18.62/20 — Ranked 1st** student in Bachelor's program

RESEARCH EXPERIENCE

Visiting Researcher

Apr. 2026 – Aug. 2026

University of Sheffield, School of Computer Science

Sheffield, UK

- Investigated **credibility bias in ESG language models**, focusing on how geographic, institutional, and linguistic factors influence automated sustainability assessment
- **Supervisor:** Dr. Nafise Sadat Moosavi (n.s.moosavi@sheffield.ac.uk)

Researcher and Data Analyst

Nov. 2020 – Aug. 2021

Iran's National Elites Foundation (INEF)

Tehran, Iran

- Served as **data-driven reliability-centered maintenance researcher** within the data analytics research group, improving maintenance of critical energy infrastructure
- **Supervisor:** Dr. Alireza Fereidunian (fereidunian@kntu.ac.ir)

PROFESSIONAL EXPERIENCE

Strategic Planning & Performance Evaluation Specialist

Jan. 2022 – Nov. 2023

Sepahan Battery Industrial Complex

Isfahan, Iran

- **Website:** <https://sepanbatter.com/en/company-overview/>

Business Development Consultant

Apr. 2021 – Nov. 2023

Karoo Behine Sanat Riva (KBSr) Company

Isfahan, Iran

- **Website:** <https://kbsr.ir/en/home/>

TEACHING EXPERIENCE

Teaching Assistant — Linear Algebra

Sept. 2017 – Feb. 2018

Kashan University

Kashan, Iran

- **Supervisor:** Dr. Mohammad Taghi Rezvan (rezvan@kashanu.ac.ir)

PUBLICATIONS

Journal Papers

- Anaraki, S. A. M., Croce, D., & Basili, R. (2025). Large language models for sustainability reporting: A systematic review and research agenda. *Sustainable Futures*, 10, 101494. doi.org/10.1016/j.sftr.2025.101494
- Fallahi, A., Mousavian Anaraki, S. A., Mokhtari, H., & Niaki, S. T. A. (2022). Blood plasma supply chain planning to respond COVID-19 pandemic: a case study. *Environment, Development and Sustainability*, 1-52. doi.org/10.1007/s10668-022-02793-7
- Anaraki, S. A. M., & Haeri, A. (2022). Soft and hard hybrid balanced clustering with innovative qualitative balancing approach. *Information Sciences*. doi.org/10.1016/j.ins.2022.09.044
- Mousavian Anaraki, S. A., Haeri, A., & Moslehi, F. (2022). Generating balanced and strong clusters based on balance-constrained clustering approach (strong balance-constrained clustering) for improving ensemble classifier performance. *Neural Computing and Applications*, 1-17. doi.org/10.1007/s00521-022-07595-6
- Mousavian Anaraki, S. A., Haeri, A., & Moslehi, F. (2021). A hybrid reciprocal model of PCA and K-means with an innovative approach of considering sub-datasets for the improvement of K-means initialization and step-by-step labeling to create clusters with high interpretability. *Pattern Analysis and Applications*, 24(3), 1387–1402. doi.org/10.1007/s10044-021-00977-x
- Mousavian, S. A., Haeri, A., & Moslehi, F. (2021). Providing a Hybrid Clustering Method as an Auxiliary System in Automatic Labeling to Divide Employee into Different Levels of Productivity and their Retention. *Iranian Journal of Management Studies*. [doi:10.22059/ijms.2021.299705.674004](https://doi.org/10.22059/ijms.2021.299705.674004)

Conference Papers

- Mousavian Anaraki, S. A. M., Croce, D., Costa, R., Tiburzi, L., Calabrese, A., & Basili, R. (2026). UniTor ESG Insights: An Interactive Human-in-the-Loop System for Traceable Sustainability Report Analysis. CIKM '26. ACM. **Accepted for demonstration.**
- Mousavian Anaraki, S. A., Croce, D., Costa, R., Tiburzi, L., Calabrese, A., & Basili, R. (2026). Unsupervised GRI-TCFD Alignment with LLM-Assisted Validation for Climate Disclosure and Greenwashing Risk Analysis. In *Proceedings of the 2nd Workshop on Ecology, Environment, and Natural Language Processing* (pp. 15–25). ELRA. doi.org/10.63317/4pebkuscwua5
- Borazio, F., Anaraki, S. A. M., Rai, S. I., Croce, D., & Basili, R. (2026). UniTor at PFB: Constrained Agentic Prompting and Self-Consistency for Financial Multi-Choice QA. In *EVALITA*.
- Anaraki, S. A. M., Croce, D., & Basili, R. (2025, September). Automatic GRI-SDG Annotation and LLM-Based Filtering for Sustainability Reports. In *Proceedings of the Eleventh Italian Conference on Computational Linguistics (CLiC-it 2025)* (pp. 775-784). <https://aclanthology.org/2025.clicit-1.73/>
- Anaraki, S. A. M., Croce, D., & Basili, R. (2025, July). Unsupervised Sustainability Report Labeling based on the integration of the GRI and SDG standards. In *Proceedings of the Fourth Workshop on NLP for Positive Impact (NLP4PI)* (pp. 151-162). aclanthology.org/2025.nlp4pi-1.13

HONORS AND AWARDS

- **Outstanding Paper Award**, Workshop on NLP for Positive Impact (NLP4PI @ ACL 2025) — July 2025
- **Ranked 1st** student in Master's program — July 2020
- Eligible to continue **directly to the Master's Program** without the National University Entrance Exam – Sept. 2019
- **Ranked 1st** student in Bachelor's program — July 2018

RESEARCH INTERESTS

- Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL)
- Information Extraction and Natural Language Processing (NLP) Applications
- Interpretability and Analysis of NLP Models
- NLP and LLMs for Environmental, Social, and Governance (ESG) Sustainability Reporting
- AI-Driven Decision-Making in Management Applications
- Operations Research Applications

SKILLS

Programming & Data: : Python, R, SQL Server

Data Visualization / BI: : Tableau, Power BI, VOSviewer

Cross-Functional Collaboration: : Ability to work effectively in interdisciplinary and multidisciplinary teams, connecting different departments and translating diverse perspectives into a shared understanding across projects

Languages: : English (Upper Intermediate), Persian (Native)